

Case 28: Overnight spread-rate adjustment

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE28_OVA. This case verifies coupling among the simulation clock, solar burn-period calculation, and surface-fire diurnal multiplier. With uniform fuel, moisture, wind, and slope, the question is whether ELMFIRE reduces and restores radial advance at the prescribed clock hours, each within 1 hour. The case exposes clock offsets, failed midnight wrapping, reversed interval tests, or an unapplied multiplier; it does not calibrate the multiplier.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For simulation time t seconds and forecast start H_0 , ELMFIRE uses

$$H(t) = \text{mod}(H_0 + t/3600, 24). \quad (1)$$

The solar routine calculates sunrise and sunset from the lower-left coordinate, year, and hour of year. With default center fraction $f_c = 0.667$ and burn length $L_b = 10$ hour,

$$H_c = H_{\text{rise}} + f_c(H_{\text{set}} - H_{\text{rise}}), \quad (2)$$

$$H_b = H_c - L_b/2, \quad H_e = H_c + L_b/2. \quad (3)$$

The stop hour is wrapped above 24. For this non-wrapping interval,

$$D(t) = \begin{cases} 1, & H_b \leq H(t) \leq H_e, \\ 0.1, & H(t) < H_b \text{ or } H(t) > H_e. \end{cases} \quad (4)$$

The spread-rate calculation multiplies its unadjusted rate by $D(t)$. Uniform flat, wind-free conditions produce an approximately circular fire whose equivalent radius $r(t) = \sqrt{A(t)/\pi}$ has full and reduced slope plateaus.

3 Derivation of expected behavior

The controlled response specifies burn-period start 7.5 hour and stop 17.5 hour. Full-to-reduced and reduced-to-full transitions are therefore expected at those respective clock hours. Since $H_0 = 10$ hour,

the events first occur approximately 7.5 hour and 21.5 hour after ignition and repeat during the three-day run. This oracle follows from the controlled targets and clock equations; calculated times are independently inferred from the arrival raster.

For the generated astronomical inputs, local sunrise and sunset are approximately $H_r = 4.5$ h and $H_s = 16.5$ h. The operational cutoff with fractional daylight $f_c = 0.667$ is

$$H_c = H_r + f_c(H_s - H_r) = 4.5 + 0.667(16.5 - 4.5) = 12.504 \text{ h} \simeq 12.5 \text{ h}.$$

A ten-hour operating window centered on H_c therefore has bounds $H_c \pm 5 = \{7.5, 17.5\}$ h. Relative to the 10:00 forecast start, the first reduction occurs after $(17.5 - 10)(3600) = 27000$ s; after midnight wrapping, full operation resumes after $(24 - 10 + 7.5)(3600) = 77400$ s. The transition metrics compare detected changes with these reference times using the tabulated timing tolerance.

4 Verification metrics and acceptance criteria

The evaluator selects the lexically latest `time_of_arrival*` GeoTIFF, masks nodata and values outside $0 \leq t < 10^8$ seconds, and requires 50 burned cells. Analysis ends at the earliest domain-edge arrival so boundary truncation cannot mimic slowdown. At 30 minute intervals, burned-cell count is converted to equivalent radius and differenced. After radius exceeds three cells, rates below half the 80th percentile of positive rates are reduced. Median full-to-reduced and reduced-to-full clock hours are then calculated.

Metric	Target	Acceptance rule	Role
Reduction starts	17.5 hour	absolute error ≤ 1 hour	burn-period stop
Reduction ends	7.5 hour	absolute error ≤ 1 hour	burn-period start

Both rows are required. A missing raster, insufficient burned area or duration, or undetected transition produces NOT EVALUABLE, never PASS.

5 Simulation configuration and predicted outcome

Input	Value	Verification role
Domain/grid	126×126 ; 30 m	circular-front resolution
CRS	EPSG:32635	location for solar calculation
Fuel/terrain	uniform FM3; zero slope	constant baseline spread
Wind/dead moisture	zero; 6, 7, 8 percent	constant forcing
Live moisture	herb 60; woody 90 percent	constant forcing
Ignition	(1890, 1890) m at $t = 0$	centered radial growth
Time control	1 s; 259 200 s stop	three daily cycles
Calendar	year 2025; hour 2890; start 10	clock coupling
Diurnal control	enabled; factor 0.1	two rate plateaus
Output	final TOA, spread rate, FLIN	case-local evidence

The predicted arrival field is concentric; contour spacing contracts overnight and expands during the burn period, with both transitions inside their acceptance bands.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

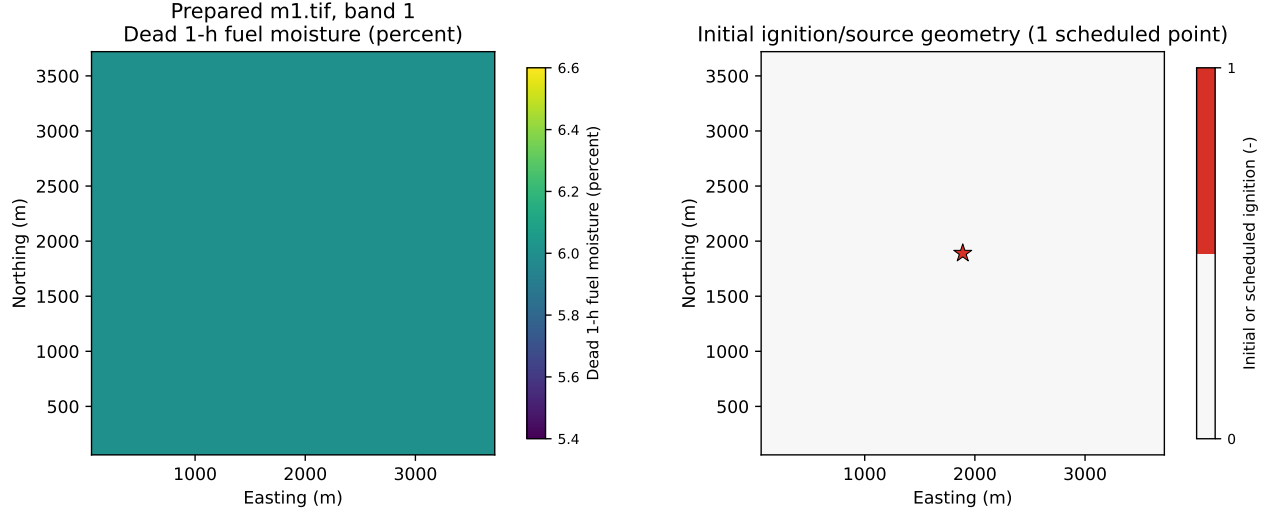


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The current artifact set contains no case-local metrics JSON or whole-domain result figure. Accordingly, the displayed status is **PASS**; 2 of 2 required rows pass across 1 completed variants. Missing measurements remain NOT EVALUABLE and historical Guide results are not substituted.

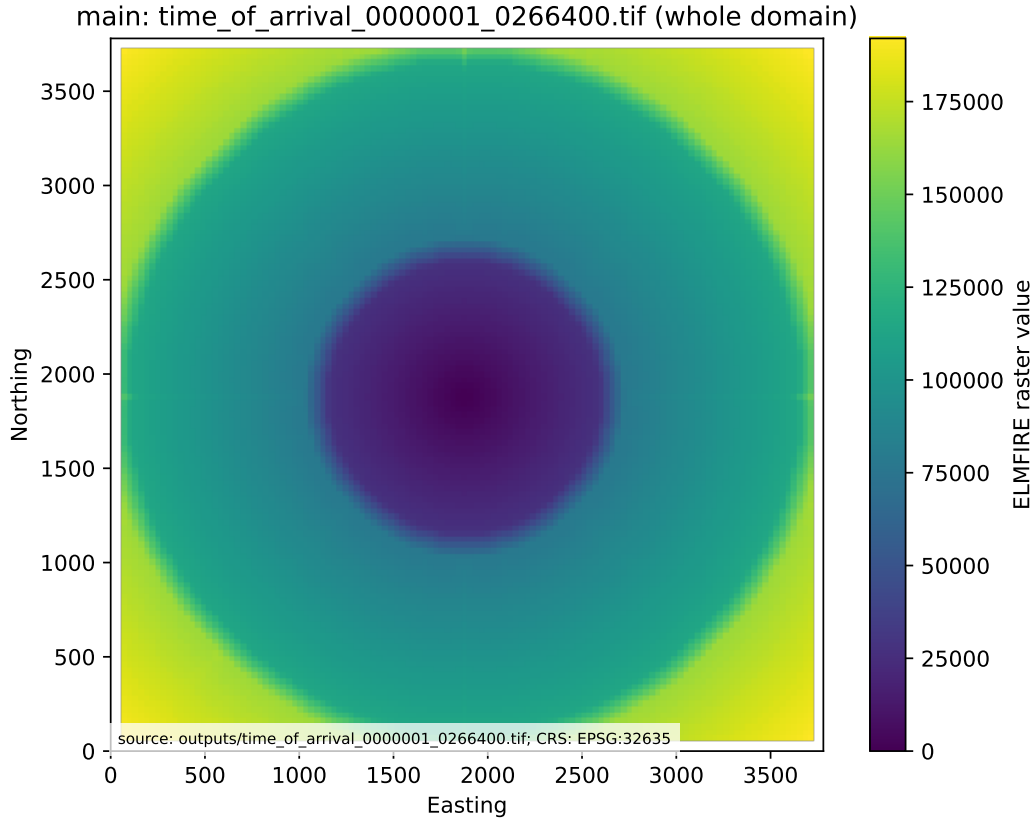


Figure 2: Final whole-domain time of arrival, in seconds, when a case-local result figure has been generated. The figure must identify its selected GeoTIFF and variant; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / reduction starts (hr)	17.5 hr	17.75 hr	PASS
main / reduction ends (hr)	7.5 hr	7.75 hr	PASS

When `outputs/metrics.json` exists, `scripts/postprocess.py` generates this table and **PASS**; when it is absent, the wrapper renders NOT EVALUABLE and no metric values are invented. Comparison symbols remain in LaTeX math mode, so $\leq$ and $\geq$ cannot be misparsed as table markup.

8 Technical interpretation

No case-local metrics artifact is currently available, so the report correctly remains NOT EVALUABLE and no timing claim is made. Once complete outputs exist, transition errors within the one-hour bands would support clock, solar-window, and diurnal-factor coupling; a common shift would implicate clock, calendar, coordinate, or solar timing, while missing contrast would suggest an unapplied multiplier or insufficient pre-edge history.

9 Scope and limitations

This synthetic case covers one date, location, fuel, factor, and grid. It does not validate factor magnitude, other seasons or coordinate systems, a burn interval wrapping midnight, heterogeneity, wind, or crown-fire, spotting, suppression, and interpolation interactions. The estimator assumes circular growth, discards post-edge behavior, samples every 30 minute, and infers rather than records the internal factor. This is implementation verification, not validation of real-fire nocturnal behavior.